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Navigating the Measurement Frontier: New Insights Into Small Farm Realities

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ABSTRACT

Measurement is not only a way of describing complex realities; it can also transform those realities by influencing policies. We live in an era of measurement innovation: new methods to deploy and new ways of adapting familiar, proven strategies to new contexts. This paper explores how new measurements provide fresh insights into the circumstances of small-farm households worldwide and describes challenges that these techniques have yet to overcome. Because the small farm sector plays a crucial role in global food security, global value chains, and rural livelihoods, understanding its conditions is a persistent focus of policymakers and researchers. I discuss how measures including satellite-based assessments of crop yields, tree cover, temperature, and rainfall, laboratory measures of soil and agricultural input quality, GPS-based plot area calculations, labor activity trackers, and high-frequency household surveys conducted via cellular phones are providing an improved understanding of fundamental dimensions of small farms and agrarian households. I identify important gaps in what is currently measured, discuss challenges related to implementing and interpreting new measures, and argue that new measurement strategies should be combined with continued investment for traditional "analog measures"—the household and farm surveys that remain fundamental for data collection in low- and middle-income countries (LMICs).

JEL Classification: Q12, Q18, C81, O13

1 | Introduction

Fundamental to economic theory and empirical analysis, sound traditional measurement strategies and plausible new ones extend the research frontier, enriching our understanding of problems and improving our means to address these challenges. Because economists rely on timely and accurate descriptions of economic growth, poverty, human capital, natural capital, and market functioning, the data we collect and how we collate and understand them are essential. Such measurement strategies can improve not only through technological advances but also through cost reductions without compromises in accuracy: granular insights acquired broadly, simply, and frequently. We study soil quality, crop production, land area, weather and its associated shocks and disruptions, prices, distances, and infrastructure quality. We collaborate with crop scientists and agronomists; we pull data from government extension services and marketing boards. Our appetite for new and improving measures opens paths to economic insights but also brings methodological responsibilities, econometric puzzles, and interpretive complications.

This paper reviews recent advances in measures related to analysis of the small farm sector in low- and middle-income countries (LMICs). I discuss areas of small farm sector research where existing measures are limited or in need of further innovation. I stress the importance of returning to the question of why and what we measure, of grounding our measures and measurement investments in strong research questions. I discuss the

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responsibility of the careful evaluation of a new measure developed outside of economics but incorporated into our discipline. I conclude by working through some opportunities and puzzles that new measures present. The contribution of this article is to present and discuss the frontier of measurement across multiple domains relevant to research into the small farm sector in LMICs. I emphasize the distinct socio-economic, environmental, and technological contexts that shape data availability, quality, and utility in these regions. Furthermore, I discuss how such advancements can foster more accurate assessments of productivity and household welfare at the farm level, while also addressing critical gaps in our understanding of smallholder dynamics, such as labor patterns, market access, and risk management.

Agricultural development economists focus on the circumstances and dynamics of the agrarian sector in LMICs. Increasing crop yields while reducing poverty and food insecurity is a long-standing goal of many governments. This goal has become increasingly urgent as countries also deal with increasing environmental and population pressures. Nonetheless, recent research suggests that crop yields in Sub-Saharan Africa are not increasing (Wollburg et al. 2024) while food insecurity is on the rise (FAO 2024). These concerning trends provide urgency and consequence to the study of long-standing questions in agricultural development.

Agricultural economists studying the small farm sector in LMICs focus on regions and production environments often characterized by incomplete, imperfect, or missing markets. This persistent engagement with systemic market failures distinguishes us from the broader community of agricultural economics.

Agricultural economists studying the small farm sector in LMICs also focus on the development challenges associated with agriculture and rural development: agricultural productivity and technology adoption, agricultural input and output market functioning and market access, land tenure and rural labor markets, and policy design and evaluation. This focus on the agricultural sector distinguishes us among the community of development economists, who are more abstractly concerned with factors that contribute to economic growth and poverty reduction.

Production function estimation and total factor productivity (TFP) analyses are workhouse methods used by agricultural economists to analyze production and allocative efficiencies and to understand how efficiently inputs are being used in the production process (Griliches 1963; Mundlak et al. 2012; Hayami and Ruttan 1970). These methods require precise measures of agricultural output, and careful measures of all inputs into production: land area under cultivation; labor hours and costs, costs of inputs including fertilizers, seeds, and agri-chemicals; investments in machinery and other farm capital; and measures of additional factors including temperature, soil quality, rainfall quantity and timing, and market prices. Collectively, these measures enable economists to discern the efficiency gains or losses over time and identify the underlying drivers of agricultural productivity changes (Lobell et al. 2009; Ruttan 2002).

Other important research questions and their associated analytical approaches in our discipline demonstrate the value of an array of high-quality measures linking agricultural production data with measures of small farm household welfare and demographics. Among these are estimations of poverty dynamics and poverty traps (Barrett and Carter 2013; Carter and Lybbert 2012); of agricultural household models linking household production and welfare (Glewwe 1991) of impacts of agricultural production practices on natural capital and ecosystem services; of the effects of shocks on agricultural production and household welfare (Arouri et al. 2015), the effects of agricultural productivity growth on poverty rates (Christiaensen et al. 2011; De Janvry and Sadoulet 2010) and of the impact of agricultural policies on farm prices, production outcomes, and household welfare (Anderson et al. 2010; Timmer 1986). New research questions in the field have also pushed the development of new measures, including women's empowerment (Alkire et al. 2013; Malapit et al. 2019) and small farmer resilience (Alloush and Carter 2024; Barrett et al. 2021).

In fact, the incomplete, missing, and imperfect markets that define the environment of small farmer decision-making require agricultural economists to take more variables into account. For example, the nonseparability of household consumption and production in smallholder farm contexts means that a host of variables related to household consumption can impact household production decisions and vice-versa (Benjamin 1992; LaFave and Thomas 2016).

It is easy to see why measurement challenges are numerous and significant for agricultural economists working in LMICs on the small farm sector. Farms are often small and fragmented; land and labor markets are largely informal; farmers tend to still heavily rely on family labor for production and harvesting; and household expenses and market access as well as input and output prices can be characterized by important within-year and within-season variability. When farmers are surveyed for information about investment, revenues, and production, their recall can be uncertain, based on undocumented or poor record keeping with regard to production costs, expenses, and sales.

The paper begins with a review of important terminology: observable variables and latent constructs, objective and subjective measures, and a brief discussion of measurement error. Section 3 is divided into subsections reviewing advances in measurement relevant to the small farm sector in a range of fields. Each subsection describes recent innovations as well as current challenges and limitations specific to the measures. Section 4 takes a cross-cutting view of the current state of measurement, working through questions relevant to interpretation and best practices of new measures. Section 4 also considers cases in which multiple measures of a quantity (objective and subjective measures, for example) are inconsistent and discusses conditions in which such discordance may be informative. Section 5 concludes the paper.

2 | Terminology

2.1 | Variable Measurement

Two distinctions are helpful regarding the kinds of concepts that agricultural economists measure. To begin with, we distinguish between **observable variables** versus **latent constructs**.

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Observable variables are those that can be directly measured using tools or observation: weight, distance, and age. Latent constructs are concepts that are not so readily quantified—theoretical concepts that we cannot measure directly and must use proxies to assess: food security (Smith and Subandoro 2007; Headey and Ecker 2013; Pinstrup-Andersen 2009), socioeconomic status and economic inequality (Ravallion 2020), risk preferences (Charness et al. 2013). Researchers use observable variables to measure latent constructs: income to proxy for socioeconomic status, for example.

The second distinction: observable variables—including those being used to measure a latent construct—can be measured using **objective measures or subjective measures**. Objective measures tend to be transparent and reproducible: for instance, temperature, water pH, and land area. Subjective measures are based on self-reports gathered in surveys: perceptions, interpretations, and opinions. Even so, not all survey-based measures are subjective; demographic and education data, for example, are objective measures often elicited using household surveys. An observable variable can be measured both objectively and subjectively (Douglas 2009; Porter 1996).

A latent construct poses special measurement challenges related to the translation of an abstract concept into a concrete measure. Many latent constructs have multiple plausible measures; some can be context-specific, while others may have several relevant dimensions. Poverty, for example, can be measured using a range of proxies: household asset wealth (Filmer and Pritchett 2001), household expenditure or income data (Deaton and Zaidi 2002), or composite indices that include some combination of assets, infrastructure, education, and health (Alkire et al. 2015). Assessing and interpreting a range of measures for a given concept poses its own interpretive opportunities and challenges. As abstract concepts, latent constructs can be characterized by a range of conceptualizations and interpretations—across researchers, across respondents, and between a researcher and respondents.

Operationalizing a latent construct means translating its abstract definition into an observable measure—either objective or subjective. Food insecurity, for example, can be measured based on an individual's objectively measured caloric intake (though this is costly and uncommon as a measurement practice). Food insecurity can also be measured using one of a host of subjective measures based on an individual or household's experience including the coping strategies that the household reports having employed (the reduced coping strategies index, or rCSI), the household's reported access to and intake of food (the Food Experience Scale—FIES) or the household's reported consumption of a range of foods over a designated time period (the food consumption score—FCS—or the household dietary diversity score—HDDS) (Maxwell et al. 2003; Cafiero et al. 2018; Wiesmann et al. 2009; Kennedy et al. 2010; Leroy et al. 2015). Several other latent constructs have special relevance for agricultural research in an era of environmental change: sustainability, ecosystem services, and environmental degradation. These are important concepts to track and measure, but they are also perceived as multi-dimensional and nuanced, and their measurement has long frustrated researchers and policymakers (Parris and Kates 2003).

2.2 | Measurement Error

A measurement is always an inference, and both objective and subjective measures can be characterized by error, differences between the true value of a variable and the measured value (Wooldridge 2008; Carletto et al. 2021). Measurement error is a problem for the internal validity of an analysis. Critical variables for estimation of agricultural production functions, for instance, are always troubled by inaccuracy: labor inputs by plot or crop, agricultural wages, labor quality, input quantity and quality, and measurement error remain an important source of bias in agricultural production functions.

Random measurement error, or "classical" measurement error, is noise—mean zero and uncorrelated with true values (in a regression model framework—uncorrelated with true dependent and independent variable values and uncorrelated with the equation error). Classical measurement error can result from factors including mis-calibrated instruments (objective measures), recording errors, or variable question interpretation by respondents (subjective measures). Classical measurement error in a regressor biases the estimator toward zero (attenuation bias) and in the dependent variable increases standard errors. The presence of random measurement error increases statistical noise, pushing up the sample sizes required for analyses. In contrast, nonrandom, nonclassical measurement error is a measurement error that exhibits some relationship between, for example, the errors in measurement and the variable's true value. Biases in parameter estimation due to nonclassical measurement errors can mislead researchers and policymakers. Recent analyses of the implications of nonclassical measurement error in agricultural data and analyses include Abay et al. (2019) and Carletto et al. (2013).

Improved measurement can ex ante reduce both random and nonrandom measurement errors (rather than using ex post statistical techniques to address the problem). Improving measurement to reduce error means increasing the precision and accuracy of our objective, instrument-based measures or improving the clarity and quality of our survey-based measures. Defining the measurement error of a measure of a latent construct involves assessing the reliability and suitability of its observable proxies.

3 | Recent Methodological Advances to Measure Small Farm Realities and Dynamics

Measurement has experienced profound changes over time, with gradual shifts punctuated by periods of significant evolution: improvements in time-keeping and associated ability to determine longitude transformed navigation in the 18th century; the French Revolution origins of the metric system; the international adoption of the metric system over the course of the 19th century and associated steps forward in standardizing modes of measurement (Klein 1974 provides a history of measurement and measures related to length, area, volume, weight, temperature, and position among others).

The current era has been characterized by new and more accurate measurement tools and declining costs for measurement and data storage. These technologies have revolutionized the study of

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important variables, including soil quality and soil moisture, temperature, rainfall, and population. In recent years, agricultural economists focusing on small-scale farming have incorporated many of these new measures into their analyses. Some work has combined new measures of variables not previously observable to farmers with randomized controlled trials or field experiments to study the way that farmers value and use new information about their soil quality, agricultural input quality, or land area, among others.

3.1 | Remote Sensing

As costs of computational power and satellite imagery access have declined, agricultural economists studying small farms have begun incorporating a range of remotely sensed data into their analyses at various spatial and temporal scales. These remotely sensed data, acquired from instruments or sensors aboard satellites, aircrafts, or drones, can provide information about crop types and land use, production volume, irrigation infrastructure and use, tree cover and biomass, plant density, temperature, rainfall, soil moisture, growing degree days and extreme heat degree days, deforestation, and land degradation (Weiss et al. 2020 and Auffhammer et al. 2013 provide reviews). Such variables are modeled from measured quantities like radiance or reflectance values, captured by sensors in various spectral bands. These data are processed using calibrated algorithms to derive measures of growing conditions and land use, facilitating research and policy decision-making in agricultural management, resource allocation, and environmental sustainability.

Because remotely sensed measures can achieve significant spatial coverage at relatively low cost, they allow researchers to monitor locations that can be difficult and expensive to reach otherwise (Wheeler and von Braun 2013). Remotely sensed measures can also refresh data with unprecedented frequency, keeping track weekly or even daily growing conditions and outcomes. Remote-sensing indices, including the normalized difference vegetation index (NDVI) and measures of solar-induced florescence, can be used to monitor crop stress and estimate crop production. Handheld global positioning system (GPS) devices have become significantly more affordable and easier to use in recent years. The costs and complexity, therefore, of mapping plots, of adding latitude and longitude coordinates to other information about houses and plots, and of linking households with remote sensing data have also decreased considerably.

3.1.1 | Geospatial Data on Environmental Conditions

Geospatial data have allowed rapidly updated spatially granular measures of conditions relevant to agricultural production, harvesting, and marketing: temperature, rainfall, and soil moisture, and have consequently reduced omitted variable bias in the analysis of farmer decision-making and investment. A special benefit of geospatial data on environmental conditions is improved awareness of the incidence and severity of pest and weather shocks. Information on these unexpected hazards for households, villages, and regions open new methods to address longstanding questions about the near- and longer-term effects of shocks for agricultural production, infrastructure and markets, and household food security (Quinn et al. 2018; Varshney et al. 2015). Improved measures of agricultural risks have also been essential to the design and evaluation of index insurance for small farmers (Carter et al. 2017; Benami et al. 2021).

3.1.2 | Satellite-Based Crop Production Estimation

Crop cuts are the gold-standard method for estimating crop production and yields in small farms: mature crops on randomly determined areas of a selected plot are harvested, dried, and weighed by researchers to estimate yields per hectare. In such studies, researchers have shown that farmer reports tend to exhibit nonclassical measurement errors in comparison with crop cut-based measures, with reporting errors often correlated with the size of a field and its associated production (Desiere and Jolliffe 2018; Gourlay et al. 2019). Satellite-based yield estimations have shown effectiveness for regions where crop production is large-scale and relatively uniform (e.g., United States, Brazil) (Cai et al. 2019; Guan et al. 2016). These methods rely on NDVI to model biomass or on measurements of solar-induced fluorescence (SIF) of chlorophyll as a proxy for plant photosynthetic activity. Promising research is underway to apply these remote-sensing-based crop production models to small farms in LMICs (Burke and Lobell 2017; Lobell et al. 2019).

3.1.3 | GPS-Based Plot Area Measures

Land area can be objectively measured using compass-and-rope (Carletto et al. 2017), an accurate but time-consuming process that remains the gold-standard method. For many years, economists working with small farmers have instead relied on reports from farmers of their plot size. In areas lacking land registration, taxation by land area, and independent surveyors, farmer-reported plot sizes may be subjective. Plot area measurement based on handheld GPS units has been shown to compare favorably with compass-and-rope (relative to farmer reports) and to correct errors and exaggerations in farmer reports (Cohen 2019; Dillon et al. 2019; Carletto et al. 2016). GPS measurements have been integrated recently into agricultural household surveys for the measurement of household parcels and plots. This method does require enumerator training and a site visit to all plots to be measured, which can add survey costs. Comparisons of farmer-reported land area versus GPS-based measures suggest significant differences (Carletto et al. 2015; Dillon et al. 2019; Abay et al. 2019, 2023), attributable to farmer reporting errors and enumerator errors, respectively. As I discuss in the final section of the paper, an accurate objective measure may not always be the preferred choice for a researcher, depending on his or her question. Research questions concerned with farmer decision-making and behavior, for example, may glean as much insight from farmers' subjective beliefs about their land area (and yields) as from an objective area measure. In combination, the two measures can be informative about farmer beliefs, investments, and production.

3.1.4 | Remote Sensing and Poverty Measures

Daytime satellite imagery enables more granular poverty estimation than nightlight analysis, which, while useful for

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cross-country comparisons, lacks precision for within-country variation in most LMICs (Henderson et al. 2012). Researchers have begun exploring the value of remotely sensed imagery in combination with existing survey data and machine learning techniques to train models that can predict poverty at a range of spatial scales (Ayush et al. 2021; Chi et al. 2022). Jean et al. (2016), for example, use demographic health survey (DHS) survey data, Living Standards Measurement Study (LSMS) survey data from the World Bank, and daytime satellite imagery to infer poverty rates in areas where ground-level data is sparse or unreliable. This new approach is reminiscent of the small area poverty estimation work of the early 2000s, which combined census data and detailed household surveys with techniques from small area statistics to address the same problem: a lack of high-quality spatially disaggregated data on poverty in LMICs (Elbers et al. 2003; Alderman et al. 2002). In fact, Van der Weide et al. (2022), comparing small area poverty estimates in Malawi based on census data, household expenditure survey data, and estimates based on remote sensing data, find that these two approaches provide similar insights about the spatial patterns of poverty.

3.1.5 | Remote Sensing and Small Farms: Limitations and Challenges

The use of remote sensing data in the economic analysis of small farms is currently characterized by a range of challenges related to accuracy, sensitivity, and privacy protections. For example, satellite-based crop production measures must overcome a range of technical challenges related to small farms, including irregular plot boundaries and intercropping (Wineman et al. 2019; Lobell et al. 2019; Tamim et al. 2024). A second challenge is, for now, relatively limited training data for ground-truthing and model calibration (Burke and Lobell 2017). Note that the reliability of measures of soil quality based on remote sensing is similarly challenged; because we tend to lack good ground data on soil parameters at sufficiently granular spatial scale across LMICs, the models remain limited and their accuracy unestablished (Gourlay et al. 2017; Kosmowski et al. 2020).

The use of satellite imagery to estimate spatially granular rates of poverty is a second area of research characterized by both possibility and limitations. Poverty estimates based on satellite imagery are likely to provide an inaccurately static measure of regional poverty if significant variables picked up by the model in the training data are relatively fixed in time (infrastructure quality and access, housing quality including roof material, terrain, and irrigation, for example). A satellite measure may fail to capture important seasonal characteristics of poverty, spatial income disparities, or social vulnerabilities. Furthermore, data privacy concerns and as yet limited geo-referenced high-quality asset, expenditure, and consumption data to train prediction models limit broader application (Ayush et al. 2021). Remote sensing data used in concert with survey data can also be characterized by measurement error from sources: miscalibration in the measurement technology, misspecification of the modeling converting the signal from the sensor or satellite to the measured variable, and the inferred spatial or temporal resolution of the data. Despite such limitations, satellite-based poverty estimation shows promise in complementing traditional survey methods.

3.2 | Soil Quality Measures

Though soil quality is not usually directly observable by researchers or farmers, it is a crucial determinant of agricultural productivity (Barrett et al. 2010) and may influence farmer learning about the suitability and profitability of new technologies (Tjernstrom 2016). Economists have long speculated that the significant heterogeneity in production outcomes in Sub-Saharan Africa could be at least in part attributable to heterogeneity in soil quality (Gollin and Udry 2021), and agricultural economists studying the small farm sector have begun to use laboratory measures of soil nutrients to characterize spatial patterns of soil quality, to study correlations of soil quality with farmer management and environmental factors (Berazneva et al. 2018; Marenya and Barrett 2009; Gourlay and Kilic 2023), to analyze the relationship between farmer beliefs about soil quality and the objectively measured soil pH, macro and micro nutrients (Berazneva et al. 2018), and to characterize relationships among soil quality and other household socioeconomic variables (Barrett and Bevis 2015). Comparisons of soil characteristics based on farmer reporting with lab-based measures find only weak correlations (Berazneva et al. 2018; Kosmowski et al. 2020). Some recent research has studied the value of lab-based soil quality measures and associated management recommendations to farmers (Harou et al. 2022; Brezaneva et al. 2023; Tamim et al. 2024). Soil quality is multidimensional, complicating data integration and comparison across studies. A final note—spectrometers— offer the possibility of future in situ measures of a handful of important soil characteristics, including organic carbon and pH, and seem to compare well with measures from lab-based analyses (Kosmowski et al. 2020). Although units remain costly, their prices are beginning to decline, offering exciting prospects for soil measures in the coming years.

The long-observed inverse relationship between crop yields and cropped area has significant implications for policy related to support for small farms in LMICs. It also serves as an example of the way that new measures can shed light on persistent research questions in the small farm sector (Barrett 1996; Deininger et al. 2022). One longstanding explanation for the relationship has long been that it is attributable to omitted variables: specifically, that soil quality—strongly correlated with productivity—is better on smaller farms relative to larger farms (Benjamin 1995) because of differences in input and cultivation practices. However, analyses incorporating plot-level soil quality measures indicate that soil quality may not, in fact, be the answer. For example, using good soil quality data at the plot level with multiple plots per household, Barrett et al. (2010) find that soil quality measurements do not appear to account for the relationship, suggesting that a better explanation may involve factor market imperfections (Foster and Rosenzwei 2022) or measurement error (Lamb 2003; Bevis and Barrett 2020; Abay et al. 2019).

Challenges related to soil quality measurement stem from the range of available testing methods, spatial scales, and dimensions. This requires careful decision-making and interpretation by economists using measures including transparency and discussion in publications and reports. For example, new in situ testing methods including near- or mid-infrared spectroscopy scanners, can provide rapid assessments of soil nutrients by measuring the soil's absorption and reflection of different wavelengths of

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light. These scanner-based measures require calibration and have limited validation so far in many LMIC contexts. As discussed, soil analyses are also multi-dimensional, requiring researchers to make informed and transparent decisions about which dimensions to focus on and which to use as proxies for latent constructs of interest, including soil fertility.

3.3 | Measuring Quality in Agricultural Input and Output Markets

3.3.1 | Nonlabor Agricultural Input Quality

Like soil quality, nonlabor agricultural input quality is not generally observed by farmers or researchers yet essential to the productivity and profitability of farmer investment. Researchers have begun to use lab-based measures to study the quality of agricultural inputs available for purchase by small farmers: nutrients in mineral fertilizer (Hoel et al. 2024; Michelson et al. 2023; Michelson et al. 2021; Sanabria et al. 2018) concentration of the active ingredient in market-available glyphosate herbicide (Ashour et al. 2019). Low-quality inputs have consequences for agricultural productivity but also impact farmer decisions about using them in sufficient quantities or even at all (Assima et al. 2017; Bold et al. 2017). A focus on nonlabor agricultural input quality also invites a research focus into the functioning and operations of input supply chains, an important area where agricultural economists working in LMICs are beginning to extend their work (Naugler et al. 2024; Dar et al. 2024; Dillon et al. 2025).

Fertilizer has been shown in this work (summarized in Michelson et al. 2023) to be reliably of good quality in these studies, with nutrient content consistent with the manufacturing standard. Most tests documented in the literature to date are assessments of the nutrient content of urea, a single-nutrient fertilizer 46% nitrogen by weight, widely used by small farmers. In contrast, glyphosate herbicide has been found to exhibit widespread quality problems; it is prone to dilution with water (Ashour et al. 2019; Haggblade et al. 2021). Seed quality has many dimensions, including germination rate, disease presence, and seedling vigor, which are observable features of seed performance that farmers also assess in the field. Researchers are beginning to document and quantify these dimensions of seed quality (Barriga and Fiala 2020; Bold et al. 2017) using moisture measures and grow-out trials; some attention has focused on assessing varietal purity—whether a seed is in fact the variety that it is advertised to be (more on this below)—especially as the costs of DNA sequencing have come down in recent years.

Researchers have collected both lab-based measures and farmer perceptions of quality for these inputs and have documented that farmers report beliefs about input quality inconsistent with lab-based analysis. For example, farmers widely report that the quality of local-market urea is poor when laboratory results indicate that it consistently meets specified standards (see, Sanabria et al. [2017] and Michelson et al. [2023] for a discussion of this point and presentation of the evidence) and, across markets in Uganda, Ashour et al. (2019) find evidence of a correlation between the documented quality problems in the market and farmer beliefs about quality in that area, but show that these beliefs are often locally inaccurate, only partially accounting for real differences in quality.

3.3.2 | DNA Fingerprinting

Agricultural economics research has begun to use DNA fingerprinting to identify specific varieties of crops under cultivation (Stevenson et al. 2018; Yigezu et al. 2018), using seed or leaf cuttings for lab analysis. DNA profiling techniques can be used for analytical and varietal purity testing of seed lots. Work of this sort has been conducted by CGIAR as part of their efforts to document the scope and impact of their decades of research on small farm crop adoption and productivity (Stevenson et al. 2018). Research has focused so far on varietal identification in cassava (Maredia et al. 2016; Wossen et al. 2019), sweet potato (Kosmowski et al. 2019), maize (Bohr et al. 2024), and bean (Rabbi et al. 2015).

Research findings on varietal identification that includes both lab analysis of farm samples and farmer self-reports has revealed significant discrepancies between what farmers believe they are growing and the actual strains under cultivation For example, Wossen et al. (2017) find that 28% of the 2500 nationally-representative cassava farmers in Nigeria assumed they were growing local varieties when they were actually growing genetically improved plant material, while 13% thought the reverse—that they were growing improved varieties when in fact they were not. Wossen et al. show that this misreporting is a nonclassical error, correlated with a handful of farmer observables, including education level and higher self-reported access to more sources of information about crop varieties. Bohr et al. (2024) document similar patterns among maize growers in Ethiopia. These apparently widespread misperceptions underscore an important theme discussed further in the final section: farmers' beliefs can offer researchers valuable insights, even when they are wrong.

3.3.3 | Marketed Crop Quality

Crop quality is relevant for understanding prices, market structure, and market segmentation. Some research into the small farm sector has begun to use new methods including laboratory measures of moisture and aflatoxin to measure quality of marketed agricultural production including maize in Uganda (Bold et al. 2022) and Kenya (Mutiga et al. 2015) and wheat in Ethiopia (Do Nascimento Miguel 2024), watermelons in China (Bai 2025), milk in Vietnam (Saenger et al. 2014) and Uganda (Ariong et al. 2024) and groundnuts in Ghana (Magnan et al. 2021). This research is related to the implementation and evaluation of policies designed to incentivize farmers to invest in higher quality production or post-harvest practices and also to research into contractual design and compliance in agricultural value chains.

3.3.4 | Measuring Quality: Limitations and Challenges

Measuring quality in both input and output markets presents several technical and social challenges. Current research varies in the degree to which it engages and foregrounds these challenges. For example, quality in both input and output markets is generally multi-dimensional and, for many attributes of a crop or an input (color, damage, size), somewhat subjective and even locally or

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temporally variable. Quality standards, incentives, and enforcement may change along the supply chain for a given crop. Second, not all relevant dimensions of quality have objective measures; freshness and hygiene, for example, may frustrate definition and quantification. Even quality characteristics with extant objective measures (glyphosate content in herbicide, germination rate, presence of pesticide residues, aflatoxin, DNA fingerprinting) may be costly and cumbersome to obtain, especially if local labs are not equipped to conduct the measurements. Michelson et al. (2024) describe these challenges with respect to input quality measurement and analysis.

DNA fingerprinting can be challenging in LMICs due to limited infrastructure for the proper collection and preservation of biological samples, leading to contamination or degradation. A lack of trained personnel and few resources for high-quality forensic analysis increases the risk of errors, making DNA evidence less reliable. Although DNA fingerprinting is somewhat straightforward for clonally propagated crops like sweet potato and cassava and in self-pollinating crops like beans, it is considerably more complicated for open-pollinated crops like maize (Bohr et al. 2024). Varietal identification based on DNA analysis in maize is a relatively harder task as pollination and hybrid crossing occur in the field. Maize DNA fingerprinting requires a genetic reference library to identify and match tested samples with true breeder varieties; building and validating this kind archive can be expensive and time-consuming, and it remains a work in progress for many crops (Warburton et al. 2002).

In situ measures are not currently available to assess most quality parameters of relevance to research and policy. Developing and testing low-cost technologies for quality monitoring in input and output is an important area of interdisciplinary work. Analysis of discrepancies among transacting parties and dynamic change in the definition and enforcement of quality attributes are promising and crucial areas for further research.

3.4 | Advances in Survey Methodology

New data sources based on Information and Communications Technologies (ICT), sensors, laboratory methods, and satellites have special advantages when combined with household farm surveys, which continue to constitute the cornerstone of agricultural data collection in LMICs. These surveys remain critical not only for validation and for ground-truthing other methods but also for targeting policies and aid, for generating essential agricultural statistics, for measuring spatial variation and patterns over time, and for providing the substrate for research into the fundamental questions of agricultural economics.

Empirical research must consistently scrutinize its own methods. Such assessment and innovation have been underway in household survey design focused on the small farm sector. Researchers have focused on the relevant scale and method for measuring a range of critical household and agricultural variables. For example, the frontier of work in recent years using multiple methods to measure the same quantity has been the World Bank LSMS team, whose Integrated Surveys on Agriculture (LSMS-ISA) initiative (which began in 2008) integrated detailed, high quality, plot-level agricultural data with household-level data, providing insights into rural households and livelihoods and rural sector conditions in eight countries (Carletto et al. 2010).[1] As a part of this work, the LSMS-ISA team has taken the lead to conduct careful comparisons of objective (generally a "gold-standard" method) and subjective (farmer reported) measures—much of this work discussed above—of variables including crop production quantities (Lobell et al. 2020) and farm plot size (Carletto et al. 2013; Kilic et al. 2017; Abay et al. 2023).

A second area of research on survey methodology in LMICs has focused on data quality, using survey experiments (many implemented by World Bank research efforts) to interrogate the quality of data produced by enumerators in questionnaires and interviews. De Weerdt et al. (2020) explore the effects that survey design can have on data quality and associated analyses and discuss best practices (Dillon et al. 2021; Carletto et al. 2021)[2]. These studies of survey methods design have provided insights into data quality and reliability: question ordering, survey length and respondent fatigue, and survey respondent choice. This work has focused on labor statistics in household surveys (Bardasi et al. 2011; Kilic et al. 2020), the importance of survey length and respondent fatigue on data quality (Jeong et al. 2023), use of production diaries to improve household crop and investment statistics (Deininger et al. 2012), the effect of recall periods on household consumption and associated welfare estimations (Beegle et al. 2012; Abate et al. 2022), and on farmer-reported crop production (Wollburg et al. 2021).

A recent paper by Carletto et al. (2022) uses a collection of nationally representative household surveys to study the relationship between income and consumption measures. Globally, both aggregates are used to measure poverty; countries and regions vary in terms of whether they rely on income or consumption, with most middle-income and upper-income countries basing their statistics on income and poorer countries on consumption. The LSMS focus on consumption data over income measurement, in part because many of the households living in countries covered by the LSMS work in agriculture and/or the informal sector, with income streams that can be seasonal or irregular within the year and largely undocumented (Beegle 2015). Carletto et al. (2022) use survey measures of consumption from 20 surveys in 5 countries to benchmark income measures and to focus on two common sources of discrepancy between income and consumption measures: underreporting and survey design features. The researchers also analyze heterogeneity in the relationship between the calculated income-consumption discrepancy based on income composition, household and individual characteristics, and characteristics of the survey instruments. Some of this work underlies and motivates recent methods to impute poverty measures in data-scarce environments (Dang et al. 2017; Dang et al. 2019), an important and active area of research.

3.4.1 | Cellular-Phone Based Surveys

The use of cellular phone-based surveys to collect household and agricultural data is developing quickly as cellular phone networks and ownership increases in LMICs, especially among the rural poor. Driven by the need to reach households safely amidst mobility restrictions and by heightened national and international interest in monitoring household welfare during

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health and economic crises, the adoption of phone-based surveys accelerated during the COVID-19 pandemic. (Josephson et al. 2021; Kanyanda et al. 2021). An associated methodological literature on cell phone survey best practices is also providing new insight (Gourlay et al. 2021; Brubaker et al. 2021) and some grounds for caution. Abate et al. (2024), for example, use a survey experiment to show that consumption measures reported by respondents over the phone are 23% lower on average than in person in an urban Ethiopian population, though a diet-based food security measure (household dietary diversity, HDDS), was largely comparable across modalities.

A related area of research works to establish poverty rates and other relevant household welfare characteristics using mobile phone call records data. This research showed early promise (Blumenstock et al. 2015) but has been somewhat hampered by legal and logistical complications for researchers accessing proprietary data held by telecommunications companies (Steele et al. 2017).

3.4.2 | High Frequency Surveys

Because the cost of adding rounds of cellular phone-based surveys is lower than the costs of additional rounds of in-person surveys, the development of cellular-based surveys has been accompanied by increased collection and analysis of high-frequency surveys. These surveys offer several attractive features, including collecting timely, even in some cases near "real-time" data. Some of these data can help alleviate accuracy problems in annual household surveys. For example, cell phone surveys have also been used to collect high-frequency labor data (Arthi et al. 2018; Pape and Mistiaen 2018) and to compare high-frequency estimates with estimates of labor hours based on recall over longer periods. Merfeld and Morduch (2023) analyze monthly consumption data in a panel of Indian households and document that poverty is more widespread than analyses based on annual data.

3.4.3 | Labor Activity Trackers

Researchers use time-use data to describe how individuals allocate their hours between on-farm labor, off-farm activities, household labor, and other responsibilities. The accurate measurement of labor hours and labor productivity has lagged other advances, even though labor is a critical component of understanding productivity, allocation, investment, and technology adoption. Household surveys remain the primary method for the collection of data about labor in agricultural production. Some recent research has tested higher frequency labor modules via cell phones (Arthi et al. 2018), but new efforts are using sensors and smartphones. When Akogun et al. (2021) used wearable accelerometers (a Fitbit is an example) to track activity and physical effort of sugarcane cutters in Nigeria, they found that measures based on the accelerometers correlated well with firm data on worker production, and they assert that careful measures of labor effort can provide insights into the relationship between effort and productivity, or in some cases serve as a valid proxy for productivity. This finding has special relevance in the analysis of farm work without a market wage, including tasks like collecting water or other activities associated with subsistence agriculture. Zanello et al. (2017) use accelerometers to measure individuals' activity in Ghana, and Picchioni et al. (2020) in Nepal; Friedman et al. (2023) use accelerometers to measure activity and to compare physical effort levels between men and women in Malawi.

Accelerometer-based data are used to supplement or supplant survey-based measures of time use within households, but important innovation has also characterized the collection of time use data. Standard methods for time-use data collection are based on survey modules, direct observation, and time diaries. Daum et al. (2019) use an open-source smartphone application called *TimeTracker* to gather time-use data for household members in real time in a study of the effects of agricultural mechanization on the time use of the members of the adopting households. Lkhagva (2020) used a similar approach in Mongolia.

3.4.4 | New Survey Methods: Challenges

Cellular phone-based surveying has some limitations in agricultural data collection, however, given the relative remoteness and low (though improving) connectivity of many farm households, the variable cell phone ownership and access patterns within households. Samples may be biased towards less-remote and less poor households and individuals. These variable access and use patterns raise concerns about representativeness of the data collected via cellular phones, and researchers tend to rely on survey weights to make corrections. High frequency and cellular phone-based surveys may face issues related to data quality and accuracy that are distinct from in person surveys but these may depend on the content of the interview (Abate et al. 2024 is a good example); these are important areas for methods research. Considerable research is needed in welfare measurement, of consumption and income and asset measure correspondence. Smart phone-based time use collection (via an application) needs more research and survey experiments to compare effects on recall bias; the links between accelerometer-based measures of labor effort could be explored in future work, especially the links between activity and effort and the relationship between effort and disutility (Friedman et al. 2023).

4 | Cross-Cutting Challenges and Opportunities

New measurement technologies and strategies bring new challenges. Some of these are familiar, like the harmonization of scale and units, or evolving, inconsistent norms for checking the accuracy of measures incorporated from outside sources. Others are specific to these new options. I discuss four examples below.

4.1 | Evaluating and Documenting Measures Incorporated From Other Disciplines

Incorporating measures that originate in other disciplines requires special attention to and documentation of data quality as processes for validation and evaluation are not native. Especially when measurement strategies are new to agricultural economists, norms for evaluation can be unclear and evolving.

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Authors of studies using such new measurement strategies face a heightened responsibility to document the quality and reliability of their findings. Economists using measures from other disciplines must understand and follow measurement protocols from those disciplines (Josephson and Michler 2023; Michelson et al. 2023), including how to recognize and account for error in the measure itself. For example, laboratory measures of soils or nutrient content in agricultural inputs normally involve double testing a randomly selected subset of samples to characterize measurement error. Economists adopting the strategy would do well to take an additional step of testing samples across multiple labs to determine measurements of quality conditional on the given laboratory. This double-checking can be crucial if the study must involve labs where quality control is not assured. When Ashour et al. (2019) analyzed results from urea fertilizer tests performed in multiple labs as a validation of their data and urea nutrient content measures for a study in Uganda, they found significant differences in the reports of nutrient quality of tested urea depending on which laboratory performed the analysis. Michelson et al. (2023) describe a similar experience for urea samples double-tested in Kenya and the United States. Interrogating data quality in the analysis of soil samples should include thorough documentation of the sampling design, discussion of the laboratory protocols for sample preparation, and analysis following established USDA or FAO methods. Michler et al. (2022) work through a similar set of best practices for the use of remote sensing measures in combination with household survey data. Poets et al. (2020) present emerging protocols for conducting DNA fingerprinting for varietal identification.

4.2 | When Is Discordance Informative?

I have already discussed examples of observable variables with multiple co-existing measures: observable variables that can be measured both subjectively and objectively, including soil quality, crop variety, and plot area and latent constructs with multiple proposed plausible objective and subjective measures, such as poverty, sustainability, environmental degradation, and food insecurity. Recent innovations in the measurement and collection of agricultural data, with multiple data sets deploying multiple methods to measure the same variable (LSMS-ISA), have created new opportunities for comparison and analysis. In many cases, however, multiple measures of the same underlying concept are not consistent. How can we interpret such discordance? When can it be informative? And when does it merely reveal measurement error rather than useful information?

Significant variations across measures can be confusing for researchers and policymakers, especially when "gold standard" measures have yet to be agreed upon or do not exist at all. For example, food insecurity is a latent construct with considerable relevance for agricultural economics research and associated policy, and a primary household and individual welfare outcome for agricultural economists working in LMICs. Household food insecurity measures are used for humanitarian and development programs and are critical inputs into the status quo global system of food insecurity crisis assessment and prediction and associated humanitarian aid allocation, the Integrated Phase Classification System (IPC). A range of measures exist for food insecurity including the household hunger score (HHS), the dietary diversity score (DDS), the reduced strategies coping index (rCSI), the food insecurity experience scale (FIES), the household food insecurity access scale (HFIAS), the self-assessed measure of food security (SAFS), and the FCS, with new measures still being introduced and proposed. This array of extant measures is recognized as capturing related dimensions of food insecurity. Even so, this range of measure relies on either recall-based objective measures of consumption characterized by classical and nonclassical measurement error, or on subjective, experience-based measures.

Nonetheless, research has established that these measures, when correlated, can provide inconsistent pictures in the cross-section of the food security status of households and regions (Maxwell et al. 2014). Work is underway to explore if that discordance can be informative. Maxwell et al. (2014), for example, argue that because the current slate of measures tends to be used interchangeably by researchers and policymakers, such research and policy formulations are likely impaired by what they term "errors of exclusion." Kim et al. (2024) work from a hypothesis that there may be useful information in the intersection of indicators: in other words, that households classified as food insecure by multiple indicators may be experiencing deprivation more severe than what is indicated by a single measurement. Kim et al. use data from Malawi's three rounds of LSMS surveys to calculate the percentage of the population classified as food insecure based on the rCSI, the FCS, or both. The analysis also includes an asset-based poverty classification for each household (defined as the bottom 20% of the asset distribution) and the intersection of these. Only 1% of the population is food insecure based on both the FCS and the rCSI indicators, and only half of those households are also asset-poor. Most of the households identified as food insecure based on the rCSI were not in the bottom quintile of the asset distribution, and only about half were identified as food insecure based on the FCS. Kim et al. (2024) suggest that these intersections of food insecurity measures with measures of household wealth may provide contextually relevant insight, helping, for example, to distinguish between chronic and acute circumstances of food insecurity. Their study reveals an important area of future work with relevance for methods, research, and policy.

Djima and Kilic (2024) explore a related question: When and how can researchers use self-reported variables measured with error to infer or more closely approximate "true" values. Focusing on crop production quantities, they use sorghum fields for which they have both "gold-standard" crop cut measures of production and farmer self-reported measures, along with publicly available remotely sensed plot-level data on other variables (including rainfall) to calibrate a model using machine learning to predict and adjust farmer self-reports out of sample. They propose this method as a means of reducing the presence of nonclassical measurement error that has been found to plague farmer self-reports and argue that the technique can be successfully implemented if one third of plots are designated for crop-cutting in the given sample (assuming relevant dimensions of variation are sufficiently similar to their Malian sample). Their analysis suggests the power of self-reported data in combination with quality objective measures of yields and geo-referencing, and it demonstrates how good georeferenced surveys can provide training data for remote sensing-based measures.

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4.3 | Farmer Perceptions—A Special Case of Measurement Discordance?

A focus of measurement improvement in agricultural economics for the last 10–15 years has been improving the accuracy of objective measures: field size, production quantity, input quantity, and quality. As the precision and accuracy of these objective measures have improved, discrepancies between what farmers report and what researchers quantify have become more apparent and pronounced.

Research has begun taking these discrepancies more seriously across a range of emerging work, documenting and analyzing differences between objective findings and farmer beliefs about input quality, field size, production quantities, and soils. Abay et al. (2023) discuss how farmer misperceptions, as a form of measurement error, can provide a basis for distinguishing between farmer misperceptions and farmer misreporting of accurate beliefs. Their work demonstrates that farmer beliefs and farmer uncertainty can have important consequences for farmer investment and adoption. For example, because farmers may conclude that an improved input is a poor investment when its quality or varietal type is not what they believe it to be, their perceptions can impact not only input adoption but also their openness to learning. Bulte et al. (2023) show that such uncertainty about maize seed quality impacts farmer investment, especially in complementary inputs like labor. Similarly, farmers holding incorrect beliefs about the type of seed they are growing (e.g., assuming that a traditional seed is a hybrid) is likely to allocate inputs suboptimally, lowering returns to adoption as well as researchers' estimates of those returns (Wossen et al. 2019; Euler et al. 2022; Bohr et al. 2024).

Researchers are also working (Hoel et al. 2024) to test whether erroneous perceptions about input quality can be corrected and to assess the impact on small farmer decision-making when these corrections are achieved (Maertens et al. 2024). This study relates to work by Abay et al. (2023), which provides farmers with the GPS-measured areas of their plots and documents how these farmers amend beliefs about their plot size. This research can be a basis for better studies of the origins of and spatial patterns of misperceptions and their response to perturbation, the resistance to change over time, and consequent implications for farmer investment in technology and learning.

4.4 | Machine Learning-Based Measures and Methods

Machine learning methods offer new avenues for addressing nonclassical measurement errors in self-reported agricultural data, such as crop yields. For instance, combining farmer-reported measures with remote sensing and "gold-standard" plot data enables calibration models to adjust self-reports and improve data quality. Yet, these methods remain constrained by the quality of their training data and require investment in robust survey frameworks to maintain validity. Athey and Imbens (2019) published a relatively good recent review of machine learning for economists. Baylis et al. (2021) and Storm et al. (2020) focus on salient issues for agricultural economics.

Machine learning methods are a powerful tool with enormous potential to analyze the small farm sector. Even with their remarkable efficiency and computational power, these approaches remain limited by the quality and representativeness of the training data. It is imperative that the training database accurately reflects the spatial and temporal variations of the target phenomena and observational conditions, accounting for inherent noise and biases to maintain the generalizability of these methods. Regardless of the methods employed to characterize agricultural systems, continual validation remains paramount. This will require continued investment in the workhorse data collection modalities of the small farm sector: agricultural censuses, household and farm surveys, and nationally representative panels. New measurement strategies will provide the most insight if we combine them and future innovations with continued sustained investment for traditional "analog measures"—the household and farm surveys that remain fundamental for data collection, policy-making, and research in LMIC.

4.5 | What Is the Relevant Scale—Digital Extension

The appropriate spatial scale for the measurements outlined above can vary. The relevant scale for one kind of research may not always be the relevant scale for policy, nor may it be the scale beneficial for farmers. In soil sampling, for instance, a challenge lies in deciding how frequently and at what granularity to collect samples to represent the variability of soil properties within a field or region. Similarly, determining the appropriate market level for price measurement — farm gate level, local markets, regional hubs, or national trading centers — can significantly influence policy decisions and economic analyses. Addressing such challenges requires nuanced methodologies, advanced technology, and interdisciplinary collaboration to ensure that measurements accurately reflect the diverse spatial contexts inherent in the small farm sector.

Special consideration and attention should be given to the question of the relevant scale of information for farmers by researchers and policymakers. A body of work is developing that provides information about growing conditions including weather forecasts, and information about local soil deficiencies and associated remediation recommendations and evaluates whether and how farmers use that information and the degree to which it impacts investments, profits, and yields (Harou et al. 2022; Corral et al. 2020; Fabregas et al. 2024; Beg et al. 2024; Burlig et al. 2024). The increasing use of cellular phones by small farmers means that this information can be provided at a large scale and minimal cost. Plot-based soil testing of farmer plots found that information alone did not change farmer behavior but in combination with liquidity, farmers responded to soil limitations that were shared by many other farmers in their village rather than their plot's more idiosyncratic deficiencies (Harou et al. 2022) and work estimating farmer willingness to pay for tests of plots that share observable characteristics with their own plot suggests that farmers believe that they can learn from and benefit from tests of others' fields (Berazneva et al. 2024). Plot-level testing may not be the actionable scale for farmers given the paucity of information currently available at village

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or even regional scale. Nonetheless, information and associated recommendations at relatively more spatially or temporally (in the case of weather forecasting) coarse scales come with a degree of uncertainty in the measure. Research into how to communicate that uncertainty effectively and responsibly to farmers is an important area for future work.

5 | Conclusion

The historian Alder (2002, 325) wrote, "Measures are more than a creation of society, they create society." Indeed, the history of measurement and the development and standardization of measures is fundamental to the history of the origin and development of states, of warfare, of taxation, and of trade. It is also fundamental to the history of science, the history of the quantification of error, risk, and uncertainty. It is the story of improvisation, of innovation. Stigler (1996) describes the instructions from a German manual for surveyors dating from 1535. The determination of a "right and lawful rod," a unit for measuring land. The surveyors were instructed to choose a somewhat random set of 16 men exiting church. These men were to be lined up heel to toe in a line. The length those 16 men made was to be divided by 16, yielding a "rod."

There is much in that description from the 1535 surveyors' manual that we recognize: a concept of randomness, a sample, a calculation of an average. It is also an example of pragmatism, doing the best with what we have. Research and measurement in the small farm sector exhibit a similar pragmatic spirit.

The integration of measurement tools such as remote sensing, GPS-based plot mapping, and lab-based soil and input and output quality assessments has begun to transform the study of small farm agricultural economics in LMIC and to provide new insight into these important questions. Remote sensing has the potential to produce granular, cost-effective monitoring of agricultural and environmental variables, while GPS-based tools can improve the accuracy of land area measurements. Emerging applications of these tools in poverty estimation and risk assessment further enhance their value for research and policymaking. However, new measurement technologies also introduce challenges. Harmonizing scales and units, ensuring data accuracy across sources, and following validation protocols from other disciplines are critical for effective integration. For example, incorporating measures from external fields such as laboratory soil testing or DNA fingerprinting requires economists to rigorously evaluate and document measurement error across multiple labs to ensure reliability. Discordance between multiple measures of the same concept, such as food insecurity or poverty, raises questions about measurement error versus potentially valuable underlying insights. Research shows that intersecting measures—for example, comparing food insecurity indices with asset-based poverty classifications—can reveal deeper patterns of deprivation, potentially distinguishing chronic from acute conditions.

Discrepancies between farmer perceptions and objective measurements—such as input quality, plot size, and production quantity—have also gained attention, highlighting the need to distinguish between misreporting and genuine misperceptions. These gaps influence farmer decisions and the adoption of improved technologies, with implications for research and policy targeting. Efforts to improve farmer understanding through GPS-based tools or corrective information campaigns show promise but also underscore the persistence of misperceptions over time.

Finally, challenges of scale remain central to agricultural measurement. For example, determining the appropriate granularity for soil sampling or market price measurement is critical to balancing research and policy relevance with farmer-specific needs. Additionally, while digital extension services leveraging mobile technology offer scalable solutions for disseminating weather forecasts and soil recommendations, researchers must carefully communicate uncertainty to avoid misinterpretation by farmers. The convergence of these advanced measurement tools with traditional methods holds promise for addressing key questions in agricultural productivity and farmer decision-making.

Though gains have been made in terms of technological innovation, data integration, and collection modalities, significant gaps remain in the measurement of the small farm sector. For example, more methodological work is needed on livestock production and marketing, labor, the timing of agricultural investments, the collection of input and output price data, and on the spatially granular characterization of agricultural risk. We need innovation to measure informal sectors of the agrarian economy: to characterize operations of traders, wholesalers, and "briefcase buyers" who operate at the farm gate. Household and farm surveys are not well designed to capture the features and dynamics of rural markets and supply chains that characterize the evolving food system in most countries. In addition, stronger efforts are required to link data through georeferencing plots and households, and innovation is urgent now in areas related to data privacy and data quality. Though innovations can remedy existing data biases and reduce measurement errors, they can also introduce fresh complications and biases. Researchers must negotiate tradeoffs between costs and benefits of reduction in biases and errors. The degree to which a measure permits a new perspective, adds new information and clarity to an analysis, and improves the precision of an estimation should be evaluated based on the context and the question, but researchers must be vigilant and rigorous in paring measures and investment back to the research question at hand. Calculating the value of increasing accuracy can be difficult and requires clarity from the researcher about the relative importance of accuracy for targeting or for parameter estimation, for example, the relative cost of errors of inclusion versus errors of exclusion. Carletto et al. (2017) offer guidance on these and other points.

Since 2008, the LSMS-ISA surveys have provided an enormously valuable resource for research into measurement and methods. The data collected by these panel surveys are highly detailed, combining agricultural data with the collection of critical variables on rural economies and household welfare. These surveys have also introduced and standardized new data collection methods (many described in this article) and conducted within-survey experiments to test and validate new measures. These high-quality panel data—which include both rich household survey data and high quality measures of agricultural investment, land, and production, have created opportunities for new research into the important relationships between household welfare and

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agriculture (Wollburg et al. 2024; Kilic et al. 2017; Kilic et al. 2015; Slavchevska 2015; Amare et al. 2018; Palacios-Lopez et al. 2017; McCarthy et al. 2018) and new insights into fundamental questions related to agricultural development including the process and complications of structural change in African economies (Wollburg et al. 2023; Gollin and Udry 2021). It was unclear at the time of writing in early 2025 whether the ISA household panels would be preserved and maintained going forward within Resilient Futures. It was similarly not yet clear whether the Resilient Future surveys would continue to invest in collecting rich agricultural data paired with household socioeconomic information. The integration of detailed agricultural data alongside socioeconomic indicators has provided a holistic picture of how households adapt to and recover from crises; these data have enabled a deeper understanding of the complex relationships between food security, income, and household coping strategies, which are essential for designing targeted interventions and policies aimed at reducing vulnerability in the face of conflict or natural disasters. The LSMS-ISA panels should continue. It would be a great loss for the LSMS-ISA panel surveys to end.

Because measurement can be a costly undertaking and investment in improving a measure's accuracy or reliability can be significant, research outlays for these purposes should be in the service of a clear objective.

Ours is an age of measurement innovation, driven by new technologies and steep reductions in data acquisition and storage costs but also by the significance of the policy and research questions that we confront: Why, after years of decline, is the global population of acutely hungry people increasing?; What are the patterns and drivers of heterogeneous increases and in some cases declines in crop yields?; What is the role of agriculture in the process of structural transformation of rural economies in LMICs today and what are the consequences for poverty and food security? How should countries support the small farm sector?; What are the impacts of climate change on agriculture and food security? How can we move beyond small farm yields to focus on household welfare gains? The research frontier is significant and exciting, and we should continue to invest in developing and validating the right tools to answer the critical policy questions we will face.

Endnotes

[1] At the time of writing this paper in early 2025, the LSMS-ISA surveys were planning to transition into "Resilient Futures" – surveys that would integrate climate, environmental and risk-related data on households in low-income countries. One objective is that the reworking of the ISAs into Resilient Futures will help provide better data for policymakers and researchers related to households' experience of and response to crises including conflict and natural disasters.

[2] Several papers cited in this section were originally presented at the conference "Survey design and measurement in development economics" held at the World Bank in 2009 and subsequently published in a special issue of the Journal of Development Economics in 2012. A second special issue focused on survey measurement was published in 2024. Both are excellent resources.

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